

**GIRRAWEE HIGH SCHOOL
MATHEMATICS**

Year 11 Mathematics Task 1

Thursday 22nd March 2007

- Instructions: a) Write all your answers on your own paper.
b) Show all necessary working.
c) Marks may be deducted for careless or badly arranged work.

Time Allowed: 90 minutes

Question 1 (21 marks)

Marks

- a) Calculate $\sqrt{\frac{36.41 - 19.57}{23.62 - 11.39}}$ correct to four significant figures 2
- b) Classify each of these real numbers as rational or irrational;
- (i) π 1
- (ii) $\sqrt{\frac{4}{11}}$ 1
- (iii) $0.12\dot{5}$ 1
- (iv) $\sqrt[3]{64}$ 1
- c) Write 0.0000000607 in scientific notation 2
- d) Simplify;
- (i) $16^{-\frac{3}{4}}$ 2
- (ii) $12a^7 \times 6a^9 \div 9a^5$ 2
- e) Find the exact value of $\frac{x^4 z}{y^4}$ where $x = \left(\frac{2}{3}\right)^2$, $y = \left(\frac{4}{3}\right)^4$, $z = \left(\frac{8}{3}\right)^3$ 3
- f) Express the following as a fraction on its simplest form;
- (i) $0.\dot{2}\dot{1}$ 3
- (ii) $0.3\dot{4}\dot{5}$ 3

Question 2 (21 marks)**Marks**

a) Simplify;

(i) $6\sqrt{2} \times 5\sqrt{7}$

1

(ii) $\sqrt{18} + \sqrt{128} - \sqrt{242}$

2

(iii) $\frac{5\sqrt{7} \times \sqrt{3}}{\sqrt{28}}$

2

b) Expand and simplify;

(i) $\sqrt{6}(\sqrt{3} - 3)$

2

(ii) $(2\sqrt{6} + 3)(2\sqrt{6} - 3)$

3

(iii) $(3\sqrt{2} - \sqrt{5})^2$

3

c) Express with a rational denominator;

(i) $\frac{3}{5\sqrt{2}}$

2

(ii) $\frac{\sqrt{7}}{3 - \sqrt{7}}$

3

d) Show that $\frac{1}{9 - 4\sqrt{5}} - \frac{4}{2 + \sqrt{5}}$ is a rational number.

3

Question 3 (21 marks)

a) Expand and simplify;

(i) $4(x + 3) - 5(2x - 3)$

2

(ii) $(2 + 3a)^2$

2

(iii) $(m^2 - 4m + 16)(m + 4)$

2

(iv) $(x + 1)^2(x - 1)^2$

3

b) Factorise;

(i) $36 - 25k^2$

2

(ii) $x^2 + 5x - 36$

2

(iii) $a^3 + 8$

2

(iv) $4m^2 + 4m - 15$

3

(v) $x^3 + 9x^2 - 4x - 36$

3

Question 4 (24 marks)

a) Factorise and simplify;

$$(i) \frac{2p+2q}{p+q} \quad 2$$

$$(ii) \frac{ac+ad+bc+bd}{a^2+ab} \quad 3$$

b) Simplify;

$$(i) \frac{x+1}{2} + \frac{x+2}{3} \quad 2$$

$$(ii) \frac{x}{x-1} - \frac{x}{x+1} \quad 3$$

$$(iii) \frac{1}{x^3-1} + \frac{x+1}{x^2+x+1} \quad 3$$

c) Simplify;

$$(i) \frac{3a^2b}{4b^3c} \times \frac{2c^2}{8a^3} \div \frac{6ac}{16b^2} \quad 2$$

$$(ii) \frac{x^2+x-2}{x+2} \times \frac{x^2-3x}{x^2-4x+3} \quad 3$$

$$(iii) \frac{m^3+1}{6m^2} \times \frac{3m}{m^2+m} \quad 3$$

$$(iv) \frac{ab-2b^2}{6a^2b} \div \frac{a^2-4ab+4b^2}{3a} \quad 3$$

Question 5 (20 marks)

a) Solve the following equations, writing solutions in simplest exact form;

$$(i) x + \frac{x}{3} = 7 \quad 2$$

$$(ii) 2m(2m+9) = 0 \quad 2$$

$$(iii) 3a^2 + 2a - 8 = 0 \quad 3$$

$$(iv) 2h^2 + 8h + 3 = 0 \quad 3$$

$$(v) \frac{2}{x+3} + \frac{x+3}{2} = \frac{10}{3} \quad 3$$

$$(vi) |x+2| = 2x-5 \quad 4$$

b) Solve $x^2 - 6x + 3 = 0$ by completing the square. 3

Question 6 (18 marks)

a) Solve each inequation, and graph your solution on the number line;

(i) $3 - 2x < 7$ 2

(ii) $-7 \leq 5x + 3 \leq 3$ 3

(iii) $|3x - 5| > 4$ 4

b) Solve the following simultaneous equations;

(i) $2x + y = 9$ 3
 $x + y = 5$

(ii) $2x + 3y = 28$ 3
 $3x + 2y = 27$

(ii) $x^2 + y^2 = 146$ 3
 $x + 2y = 21$

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Question 1 (21)

a) $\sqrt{36.41 - 19.57} = 1.173$ (2)

b) (i) irrational (1)

(ii) irrational (1)

(iii) rational (1)

(iv) rational (1)

c) $0.000\ 000\ 060\ 7 = 6.07 \times 10^{-8}$ (2)

d) $(16)^{\frac{1}{3}} = \sqrt[3]{16}$ (2)

$= \frac{1}{2^3}$ (2)

$= \frac{1}{8}$ (2)

(ii) $12a^7 \times 6a^9 \div 9a^5$

$= 8a^{11}$ (2)

e) $\frac{\pm 1}{4} = \left(\frac{2}{3}\right)^8 \times \left(\frac{3}{4}\right)^3 \times \left(\frac{3}{4}\right)^{16}$

$= \frac{2^8}{3^8} \times \frac{3^9}{4^3} \times \frac{3^{16}}{4^{16}}$

$= \frac{2^{15}}{3^5 \times 4^{19}}$

$= \frac{2^{15}}{3^5 \times 2^{38}}$ (3)

f) a) $0.2\dot{1} = \frac{21}{99}$

$= \frac{7}{33}$ (3)

(ii) $0.34\dot{5} = \frac{345 - 3}{990}$

$= \frac{342}{990}$

$= \frac{19}{55}$ (3)

Question 2 (21)

a) (i) $6\sqrt{2} \times 5\sqrt{7} = 30\sqrt{14}$ (1)

(ii) $\sqrt{18} + \sqrt{128} = \sqrt{242}$

$= 3\sqrt{2} + 8\sqrt{2} = 11\sqrt{2}$ (2)

$= 0$

(iii) $\frac{5\sqrt{7} \times \sqrt{3}}{\sqrt{24}} = \frac{5\sqrt{3}}{2}$ (2)

b) a) $\sqrt{6}(\sqrt{3} - 3)$

$= \sqrt{18} - 3\sqrt{6}$

$= 3\sqrt{2} - 3\sqrt{6}$ (2)

(ii) $(2\sqrt{6} + 3)(2\sqrt{6} - 3)$

$= 24 - 9$ (3)

(iii) $(3\sqrt{2} - \sqrt{5})^2$

$= 18 - 6\sqrt{10} + 5$

$= 23 - 6\sqrt{10}$ (3)

c) (i) $\frac{3}{5\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{2}}{10}$ (2)

(ii) $\frac{\sqrt{7}}{3 - \sqrt{7}} \times \frac{3 + \sqrt{7}}{3 + \sqrt{7}}$

$= \frac{3\sqrt{7} + 7}{9 - 7}$

$= \frac{3\sqrt{7} + 7}{2}$ (3)

d) $\frac{1}{9 - 4\sqrt{5}} - \frac{2 + \sqrt{5}}{9 + 4\sqrt{5}}$

$= \frac{1}{9 - 4\sqrt{5}} \times \frac{9 + 4\sqrt{5}}{9 + 4\sqrt{5}} - \frac{2 + \sqrt{5}}{9 + 4\sqrt{5}} \times \frac{2 - \sqrt{5}}{2 - \sqrt{5}}$

$= \frac{9 + 4\sqrt{5}}{81 - 80} - \frac{8 - 4\sqrt{5}}{4 - 5}$

$= 9 + 4\sqrt{5} + 8 - 4\sqrt{5}$

$= 17$ (3)

Question 3 (21)

a) (i) $4(2x + 3) - 5(2x - 3)$

$= 4(2x + 12 - 10x + 15)$

$= -6(2x + 27)$ (2)

(ii) $(2 + 3a)^2$

$= 4 + 12a + 9a^2$ (2)

(iii) $(m^2 - 4m + 16)(m + 4)$

$= m^3 + 4m^2 - 4m^3 - 16m^2 + 16m + 64$

$= m^3 + 64$ (2)

(iv) $(x + 1)^2(x - 1)^2$

$= (x^2 + 1)^2(x^2 - 1)^2$

$= (x^4 - 2x^2 + 1)(x^4 - 2x^2 + 1)$ (3)

b) a) $36 - 25k^2$

$= (6 + 5k)(6 - 5k)$ (2)

(ii) $x^2 + 5x - 36$

$= (x + 9)(x - 4)$ (2)

(iii) $a^3 + 8$

$= (a + 2)(a^2 - 2a + 4)$ (2)

(iv) $4m^2 + 4m - 15$

$= 4m^2 + 4m - 6m - 15$

$= 9m(4m + 5) - 3(4m + 5)$

$= (4m + 5)(9m - 3)$ (3)

(v) $2x^3 + 9x^2 - 4x - 36$

$= x^2(2x + 9) - 4(2x + 9)$

$= (2x + 9)(x^2 - 4)$

$= (2x + 9)(x + 2)(x - 2)$ (3)

Question 4 (24)

a) (i) $\frac{2x + 2y}{p + q} = \frac{2(p + q)}{(p + q)}$

$= 2$ (2)

(ii) $\frac{ac + ad + bc + bd}{a^2 + ab}$

$= \frac{(a + b)(c + d)}{a(a + b)}$

$= \frac{c + d}{a}$ (3)

b) (i) $\frac{2x + 1}{2} + \frac{2x + 2}{3} = \frac{3(2x + 1) + 2(2x + 2)}{6}$

$= \frac{5x + 7}{6}$ (2)

(ii) $\frac{x}{x - 1} - \frac{x}{x + 1} = \frac{x(x + 1) - x(x - 1)}{(x - 1)(x + 1)}$

$= \frac{x^2 + x - x^2 + x}{(x - 1)(x + 1)}$

$= \frac{2x}{(x - 1)(x + 1)}$

$= \frac{2x}{(x - 1)(x + 1)}$ (3)

(iii) $\frac{1}{x^3 - 1} + \frac{2x + 1}{x^3 + 2x + 1} = \frac{1 + (2x + 1)(x - 1)}{(x^3 - 1)(x^3 + 2x + 1)}$

$= \frac{x^2}{(x - 1)(x^3 + 2x + 1)}$ (3)

c) (i) $\frac{3a^2b}{4b^3c} \times \frac{2c^2}{8a^3} \div \frac{6ac}{16b^3}$

$= \frac{3a^2b}{4b^3c} \times \frac{2c^2}{8a^3} \times \frac{16b^3}{24ac}$

$= \frac{1}{2a^2}$ (2)

(ii) $\frac{x^2 + x - 2}{x + 2} \times \frac{x^2 - 3x}{x^2 - 4x + 3}$

$= \frac{(x + 2)(x - 1)}{(x + 2)} \times \frac{x(x - 3)}{(x - 3)(x - 1)}$

$= x$ (3)

(iii) $\frac{m^3 + 1}{6m^2} \times \frac{3m}{m^3 + m}$

$= \frac{(m + 1)(m^2 - m + 1)}{6m^2} \times \frac{3m}{m(m + 1)}$

$= \frac{m^2 - m + 1}{2m^2}$ (3)

(iv) $\frac{ab - 2b^2}{6a^2b} \div \frac{a^3 - 4ab + 4b^3}{3a}$

$= \frac{b(a - 2b)}{6a^2b} \times \frac{3a}{(a - 2b)^2}$

$= \frac{1}{2a(a - 2b)}$ (3)

Question 5 (20)

a) (i) $x + \frac{x}{3} = 7$
 $3x + x = 21$
 $4x = 21$
 $x = \frac{21}{4}$ (2)

(ii) $2m(2m+9) = 0$
 $m = 0$ or $m = -\frac{9}{2}$ (2)

(iii) $3a^2 + 2a - 8 = 0$
 $3a^2 + 6a - 4a - 8 = 0$
 $3a(a+2) - 4(a+2) = 0$
 $(a+2)(3a-4) = 0$
 $a = -2$ or $a = \frac{4}{3}$ (3)

(iv) $2h^2 + 8h + 3 = 0$
 $h = \frac{-8 \pm \sqrt{64 - 24}}{4}$
 $= \frac{-8 \pm \sqrt{40}}{4}$
 $= \frac{-8 \pm 2\sqrt{10}}{4}$
 $= \frac{-2 \pm \sqrt{10}}{2}$ (3)

(v) $\frac{2}{x+3} + \frac{x+3}{2} = \frac{10}{3}$
 $\frac{2 + 3(x+3)^2}{2(x+3)} = \frac{20(x+3)}{3}$
 $12 + 3x^2 + 12x + 27 = 20x + 60$
 $3x^2 - 8x - 21 = 0$
 $3x^2 - 9x + 7x - 21 = 0$
 $3x(x-3) + 7(x-3) = 0$
 $(x-3)(3x+7) = 0$
 $x = 3$ or $x = -\frac{7}{3}$ (3)

(vi) $|x+2| = 2x-5$
 $x+2 = 2x-5$ or $x+2 = -(2x-5)$
 $x = 7$ or $x+2 = -2x+5$
 $3x = 3$
 $x = 1$
 Not solution (4)

b) $x^2 - 6x + 3 = 0$
 $x^2 - 6x = -3$
 $x^2 - 6x + 9 = 6$
 $(x-3)^2 = 6$
 $x-3 = \pm\sqrt{6}$
 $x = 3 \pm \sqrt{6}$ (3)

Question 6 (18)

a) (i) $3-2x < 7$
 $-2x < 4$
 $x > -2$ (2)

(ii) $-7 \leq 5x+3 \leq 3$
 $-10 \leq 5x \leq 0$
 $-2 \leq x \leq 0$ (3)

(iii) $|3x-5| > 4$
 $3x-5 > 4$ or $-(3x-5) > 4$
 $3x > 9$ or $-3x+5 > 4$
 $x > 3$ or $-3x > -1$
 $x < \frac{1}{3}$ (4)

b) (i) $2x+y = 9$ (1)
 $x+y = 5$
 $\therefore x = 4, y = 1$ (3)

(ii) $2x+3y = 28$
 $3x+2y = 27$
 $6x+9y = 84$ (1)
 $6x+4y = 54$
 $5y = 30$
 $y = 6$ (3)

(iii) $x^2 + y^2 = 146$
 $x + y = 27$
 $x = 27 - y$
 $\therefore (27-y)^2 + y^2 = 146$
 $441 - 84y + 4y^2 + y^2 = 146$
 $5y^2 - 84y + 295 = 0$
 $(5y-59)(y-5) = 0$
 $y = \frac{59}{5}$ or $y = 5$ (3)

$\therefore x = \frac{13}{5}, y = \frac{59}{5}$ or $x = 11, y = 5$